From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

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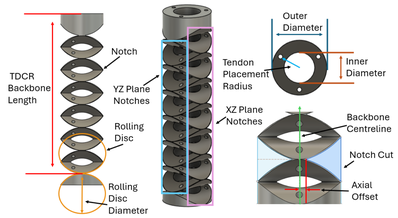
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PART A How does geometry govern compliance? — under review, ICRA 2027 (IEEE International Conference on Robotics and Automation)

PART B A sensor-free twin of one SWEEP design — in preparation for IEEE RA-L (IEEE Robotics & Automation Society)

The robot, and two questions

Printed notched tubes are cheap continuum robots for tight spaces, but there is no rule for choosing their notches and no live, sensor-free model of their shape.



TPU tube with two notch families and tendons in the wall; 160 mm modules.

A. Design Which geometry gives a wanted bend, and what stroke and force does it cost?

B. Model Can the whole shape be predicted live from servo commands alone, for any number of modules?

What is new

- **A** SWEEP: CAD to FEM, unattended; 616 runs link geometry to bend and force.
- **B** A twin from the CAD's sections, pull-only tendons and gravity: 1 to 4 ms.
- **B** It learns the module's law, not the shape: fitted on 1–2 modules, it holds on 4.

A1 SWEEP: geometry to simulation, unattended

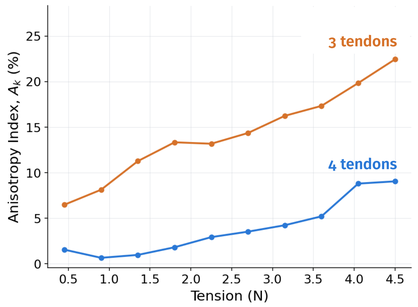
Parameters D_{RD}, L, D_o, D_i, e, c, N_t → **CAD** CadQuery, one Boolean cut → **Mesh** CGAL tetrahedra → **FEM** SOFA, pull-only tendons → **Metrics** curvature, stroke, arc fit



designs	14
tendons × loads	4 × 11
FEM runs (done)	616 (614)
TPU modulus	56 MPa
gravity	on
claims valid for	$T \le 0.9$ N

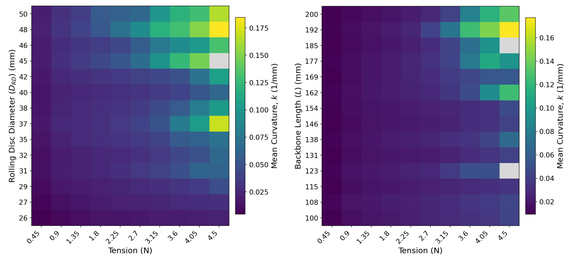
Same load, three disc diameters ($R_d = D_{RD}$): a larger disc packs more notches and bends further.

A2 Match the actuation to the structure



Three tendons bend unevenly with direction. Four cut the anisotropy $A_\kappa = \sigma_\kappa / |\bar{\kappa}|$ from 6.2 % to 2.6 % (13 of 14 designs), with two motors: antagonistic pairs.

A3 What each parameter controls



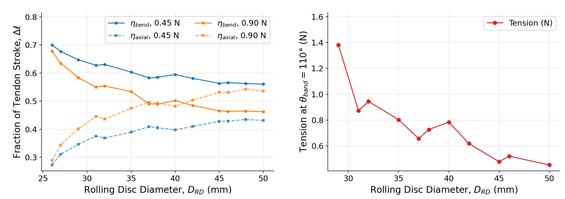
D_{RD} sets curvature turning 54° → 143° (26 to 50 mm, 0.9 N)

L sets reach tip travel 78 → 157 mm (100 to 200 mm)

Mean curvature (1/mm). More notches bend further but stray up to 5× further from constant curvature. Grey: no settled run.

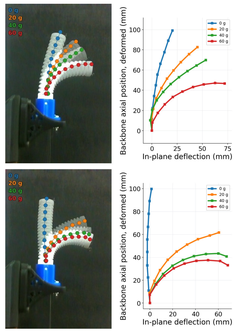
A4 Stroke budget: what a bend costs the actuator

$$\underbrace{\Delta\ell}_{\text{tendon stroke}} = \underbrace{\Theta\, d}_{\text{bending}} + \underbrace{\Delta L}_{\text{axial compression}}$$



The budget closes to 0.99; 42 to 47 % of the stroke only shortens the body (left). A larger disc halves the tension for a 110° bend, 0.87 to 0.45 N (right).

A5 Experiment: three printed prototypes



D_{RD} (mm)	NRMSE, % of length centreline	tip	axial
30	18.5	20.9	43.1
67.89	10.1	23.5	11.9
86.84	16.3	64.7	23.9

Shape agrees within 18.5 %; the tip less well (no friction or self-contact in the model). Prototypes span D_{RD} = 30 to 87 mm.

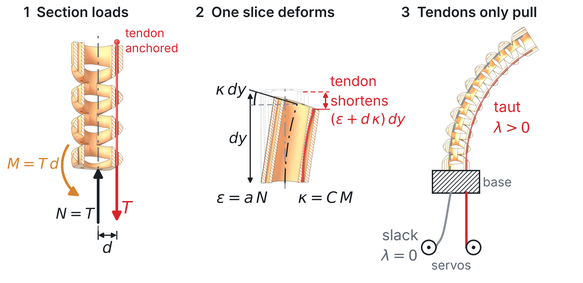
B1 Servo commands in, whole shape out

Servo commands $\delta\ell$ → **Tendon map** $\delta\ell = G\lambda$ from the CAD → **Pull-only** $\lambda \ge 0$, convex exact slack → **Gravity** fixed point CAD weight → **Shape** every 5 mm in 1 to 4 ms

its weight moves the tendon paths

No sensor on the robot. Laptop CPU, slowest call 26 ms; a Cosserat rod takes 360 ms.

B2 The physics: a beam that follows the CAD

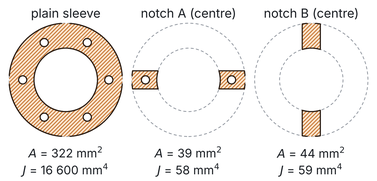


$$\delta\ell = G\lambda, \quad G_{ij} = \sum_c \tau_{ic}\tau_{jc}\left(a_c + \boldsymbol{d}_i^\top C_c\, \boldsymbol{d}_j\right) + c_t L_i\, \delta_{ij}$$

$$\lambda^\star = \arg\min_{\lambda \ge 0} \tfrac{1}{2}\lambda^\top G\lambda - \delta\ell^\top\lambda$$

Tendon map: exactly linear, any bend

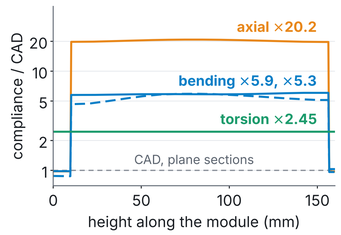
Pull-only: convex, unique, exact



Two 4 mm bridges remain: 280 times less bending stiffness, so the notches do the bending.

λ: tendon tensions; $\tau_{ic} = 1$ where tendon i passes slice c; a_c, C_c: slice compliances from the CAD; $c_t L_i$: tendon stretch.

B3 Identify the module law once, then verify



Fitted through the solver on 675 FEM poses: four scales on the CAD's compliance, and a smooth correction within 13 %.

solver check	difference
closed-form beams	< 1 µm
Cosserat rod	0.004 mm
automated tests	132 pass

B4 Scientific ML: learn the law, not the solution

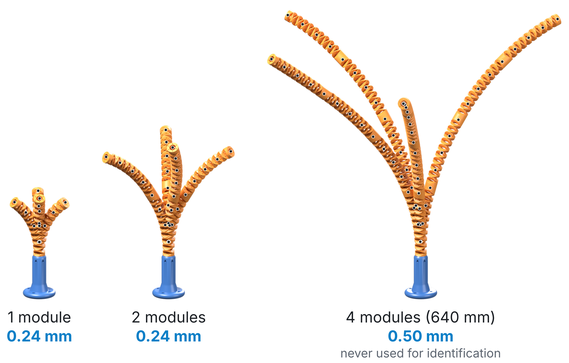
Learned law θ $C_\theta(y) = C_{CAD}(y)\, e^\theta$ 4 scales + smooth profile → **Exact physics** pull-only solve + gravity → **Shape** ⟷ loss → **Data** FEM now, mocap next

fit θ through the solver

	1 and 2 modules	4 modules (unseen)
tip error vs FEM (mm)		
learn the **shape** (physics-informed network)	3 to 6	54 to 73, retrained
learn the law, in the solver	**0.2 to 0.4**	**0.50, same θ**

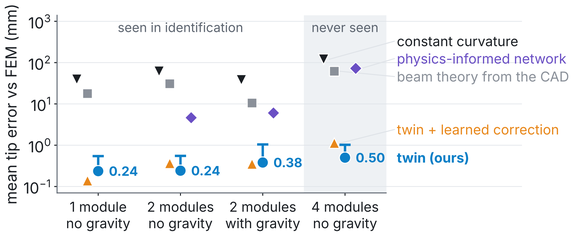
One θ for any number of modules.

B5 Calibrate small, deploy large (FEM ground truth)



1 module **0.24 mm** · 2 modules **0.24 mm** · 4 modules (640 mm) **0.50 mm** never used for identification

Fitted on one and two modules; on four, the error is **0.08 %** of length. The twin rendered on the CAD at one scale; dots: FEM centreline.



Mean and 95th percentile over held-out poses, bends to 75°. Constant curvature over-bends 1.7×: it ignores A4's compression. FEM meshes differ by 0.6 mm.

B6 Towards hardware: what the data can identify

problem (synthetic data)	remedy in the calibration
weight and stiffness confounded	weigh modules, fix the mass
0.4 mm moment-arm error	fit as 5th number, to 0.02 mm



Right: the parts designed for the four-module robot.

Conclusions

A. Design. The disc diameter sets curvature (54° → 143° at 0.9 N), the length sets reach, and 42 to 47 % of the stroke is compression.

B. Model. Identified on one and two modules, the twin predicts four to 0.50 mm (0.08 % of length) in 1 to 4 ms; a learned shape does not.

Limits. FEM only, at its resolution (about 0.6 mm); quasi-static; no friction or contact in the identification; bends up to 75°.

Builds on: Camarillo, T-RO 2008 • Rucker & Webster, T-RO 2011 • SOFA, 2012 • SoftRobots, 2017 • Kim, 2025 • Danesh, 2026.

Direction A geometry-conditioned twin for inverse design

1 Hardware validation (now)
Calibrate **5 numbers on 15 poses** of 2 modules, test on 4.
Pass: tip error ≤ 2 % of length, set before any data.

→ **2 Law as a function of geometry**
One law per SWEEP design: $\theta = f(D_{RD}, D_i, L)$.
Pass: beats beam theory on unseen designs.

→ **3 Inverse design**
Wanted bend and stroke → geometry, in milliseconds.
Pass: a printed design meets its target.

Outputs and my role

[1] D. Yadav, **O. M. Pawar,** Krishnapriya R, S. Gandhi, M. Vadali, **under review, IEEE ICRA 2027.**
[2] O. M. Pawar et al., sensor-free digital twin, **in preparation, IEEE RA-L.**

Role. Part A: SOFA 25.06 port, parallel sweeps, mesh and settling checks. Part B: model, solver, identification, calibration, baselines and software.

Thanks to Divij Yadav, Krishnapriya R and Shaili Gandhi for SWEEP and the prototypes.

Report (PDF) · Code (GitHub)